

Data Structures And Analysis

Software Requirements Specification
(SRS) Document

Project Name: Cybershooter

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1. Introduction

1.1 Purpose

The purpose of CyberShooter is to provide an interactive shooting arcade experience where the player controls a jet, shoots down enemy ships and earns points. The project aims to demonstrate practical implementation of object-oriented programming and game design principles using Python and Pygame.

1.2 Document Conventions

This document follows IEEE SRS formatting conventions. Headings are numbered, functional requirements are listed using numbered subsections, and technical terms are italicized where needed.

1.3 Intended Audience

This SRS is intended for:

- Testers: To verify the system against documented requirements.
- Project Supervisors/Professors: To review progress and compliance with standards.

1.4 Scope

CyberShooter is a 2D arcade-style shooting game developed in Python using Pygame. The scope of this software includes designing a player-controlled jet, bullet shooting mechanics, enemy spawning, collision detection, scoring and game over states. The game provides entertainment, demonstrates programming skills and serves as a foundation for future game development projects.

1.5 References

- IEEE Software Requirements Specification Standards
- Python official documentation (<https://docs.python.org/3/>)
- Pygame documentation (<https://www.pygame.org/docs/>)

2. General Description

2.1 Product Perspective

CyberShooter is a stand-alone arcade game. It was inspired by traditional space shooter games but implemented using Python data structures. It does not rely on external software beyond Pygame and runs locally on a user's computer.

2.2 Product Features

- Player controls a jet with keyboard inputs.
- Player can shoot bullets to destroy enemies.
- Enemies spawn randomly from the top of the screen.
- Collision detection removes enemies and awards points.
- High score tracking system.
- Game over screen with restart option.

2.3 User Class Characteristics

Users are casual players and students. They require basic computer literacy to use keyboard controls. No prior gaming experience is necessary.

2.4 Operating Environment

The game runs on Windows, Linux, or macOS with Python 3.x and Pygame installed. Recommended hardware: 4GB RAM, dual-core processor standard keyboard.

2.5 Constraints

- Limited graphics since Pygame is 2D.
- Performance depends on user's system speed.
- No multiplayer support in current version.

2.6 Assumptions and Dependencies

Assumptions:

- Users have Python and Pygame installed.
- Players understand basic keyboard controls.

Dependencies:

- Python 3.x
- Pygame library

3. System Requirements

3.1 Functional requirements

Functional Requirements:

- The system shall allow the player to move left and right.
- The system shall allow the player to shoot bullets.
- The system shall spawn enemies at timed intervals.
- The system shall detect collisions between bullets and enemies.
- The system shall track and display the player's score.
- The system shall end the game if an enemy collides with the player.

4. External Interface Requirements

4.1 User Interfaces

The user interacts through a keyboard interface:

- LEFT arrow: Move jet left
- RIGHT arrow: Move jet right
- SPACE: Shoot bullets

Game screens:

- Intro Screen: Title, instructions
- Game Screen: Jet, enemies, bullets, score
- Game Over Screen: Final score and retry option

4.2 Hardware Interfaces

CyberShooter requires:

- Keyboard for input
- Monitor for display
- CPU and RAM for processing

4.3 Communications Interfaces

No external communication interfaces are required.

4.4 Software Interfaces

Dependencies:

- Python 3.x for program execution
- Pygame for graphics and event handling

5. Non-Functional Requirements

5.1 Performance Requirements

The game shall run at 60 FPS on a standard PC. Enemy spawning and collision detection must respond within 50ms to ensure smooth gameplay.

5.2 Safety Requirements

The software shall not harm the user's system. It does not access files or the internet.

5.3 Security Requirements

The game shall not store personal information. Security risks are minimal since no networking is used.

5.4 Software Quality Attributes

Qualities include:

- Usability: Simple controls.
- Maintainability: Code structured in functions.
- Reliability: Stable gameplay without crashes.
- Portability: Runs on any OS with Python & Pygame.

5.5 Other Requirements

Future improvements may include:

- Power-ups and advanced levels
- Background music and sound effects
- Multiplayer support
- Enhanced graphics

6. Design / Flow chart

